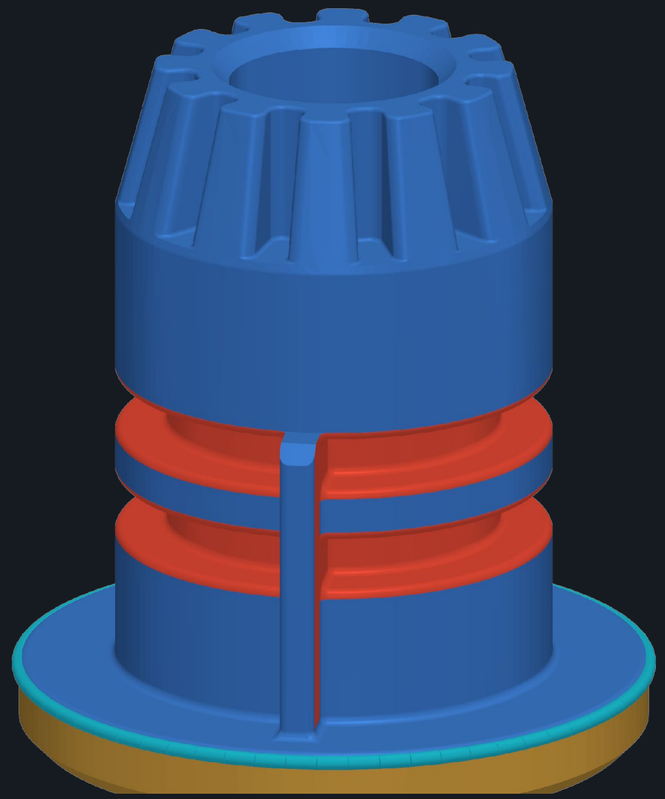
PHASE 2 · AUGUST 2026

DfM Auto-Analyzer

Automated injection-molding manufacturability analysis for STEP parts.



core cavity undercut parting line

TEAM STACK · BMS COLLEGE OF ENGINEERING

Vivek Boora Ayush Pandey Nitin Afeera

THE PROBLEM

DfM review is the queue, not the work

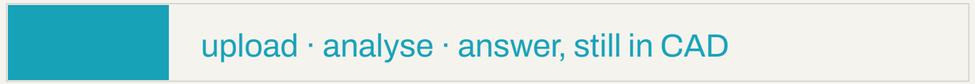
3–4 hours → under a minute

A tooling expert inspects drafts, undercuts and the split line by hand — and only once geometry is frozen. The checks are geometric and repeatable. They do not need a human every time.

MANUAL REVIEW



DFM AGENT



One STEP file in. Five decisions out.

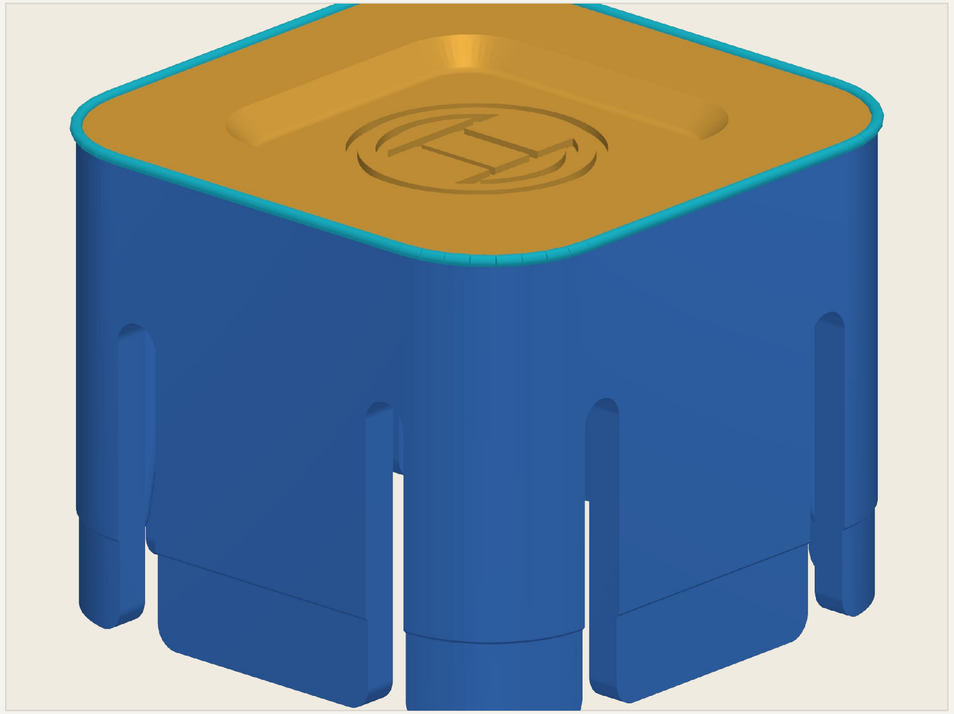
01	**Pull direction**	Ranked, runners-up shown
02	**Core / cavity split**	Which half forms each face
03	**Parting line**	One closed loop where halves meet
04	**Undercuts as features**	What is trapped, and what releases it
05	**Draft warnings + score**	Ejection risk, and one number for triage

Every answer is **geometric and deterministic** — no model, no training data, no guesswork.

THE VOCABULARY

Five words the whole tool is built on

Pull direction	The axis the mold opens along — every other answer depends on it
Core / cavity	Which half forms each face. Internal surfaces are core, even when they look like outer walls
Parting line	The closed loop where the halves meet. If it doesn’t close, that’s solid steel
Undercut	Geometry neither half can pull away from. Not a defect — a decision
Draft	The taper that lets a part eject



The same colours run through every screenshot in this deck.

THE PARTING LINE, DEFINED

A broken loop is solid steel

The rules a tool designer applies — before any of this is code.

01	**It must close.**	An open chain isn't a parting line — the mold can't separate.
02	**Every face must escape.**	Along +pull or −pull. A face neither half releases is trapped — which is why undercuts and parting lines are one problem, not two.
03	**Where physics is silent, take the shorter split.**	A wall reachable from both halves is genuinely undecided — either choice gives a working mold. We give it to the side it shares the most edge length with, because that shared edge *is* the parting line produced.

A fixed "always cavity" default routed Part 1's boundary around every snap-tab window instead of closing at the rim. Regions decided by shared edge length close it at the rim, flat, in one loop.

ARCHITECTURE & PIPELINE

Three tiers, eight geometric stages

CLIENT

React · Vite · three.js

Interactive 3D, per-face colouring, exploded halves, direction override

API

FastAPI (Python)

`POST /analyze · /analyze/directions`

GEOMETRY ENGINE

OpenCASCADE via OCP · cadquery

Sampling, raycasting, classification, parting line, regions

Regression harness `analyze_cli.py + pytest`

No per-seat licence anywhere. It runs locally, so no part geometry leaves the machine.

Each stage is a geometric test, not a heuristic. Teal stages are new since Phase 1.

01
Parse & sample
Up to 15 UV samples per face, each with its exact normal

02
Build the raycaster
One intersector, reused for every direction

03
Search the pull axis
9 fixed axes

04
Pick the sign
Set by where the internal features open

05
Split silhouettes
Both-half faces cut along the clamshell plane

06
Classify
Per face, area-weighted

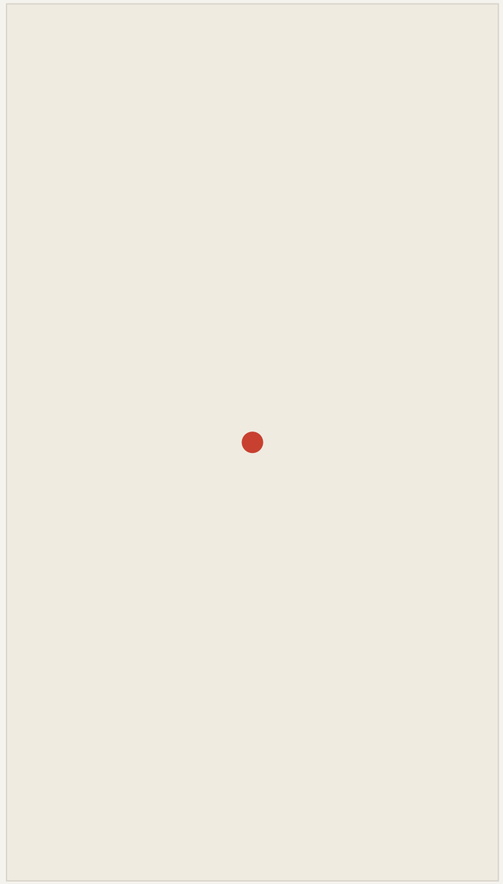
07
Extract the parting line
Chain edges, verify closure, return one loop

08
Group & score
Trapped faces → features; penalties → one number

DEEP DIVE · UNDERCUT DETECTION

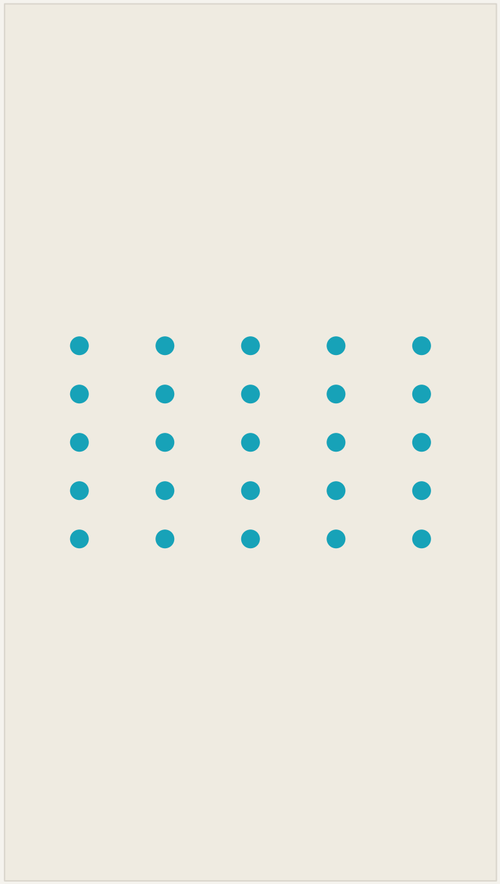
Undercuts: ask the surface, not one point

JUNE · 1 SAMPLE



4 undercuts

AUGUST · UV GRID



0 undercuts

Multi-sampling

Up to 15 UV points per face, kept only if they lie on the trimmed face. A face is an undercut when ≥ 50 % of its samples are trapped.

Transition filter

Only a hit where the ray *enters* material blocks the pull. Grazing a neighbouring fillet no longer counts.

> A face is trapped only if **neither** mold half can pull away from it.
> A vertical wall is trapped only if **both** ways are blocked.

DEEP DIVE · THE THREE DECISIONS

Direction, split, and the line between the halves

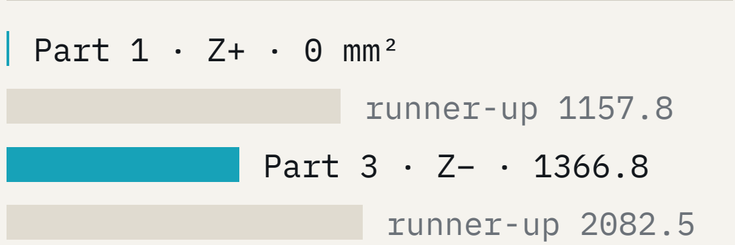
01 · PULL DIRECTION

Ranked by trapped **area** , count only as tie-break.

Axes, not directions — undercuts are symmetric in ±d.

Branch-and-bound abandons a losing axis early; a pruned axis reads **≥ N** , never an exact figure.

Override with any candidate or a typed vector — the whole analysis recomputes.



02 · CORE VS CAVITY

Clear normal — area-weighted vote on n·d.

Vertical wall — reachability; if the cavity path is blocked, only the core can form it.

Through-hole — core, by the core-pin convention.

Reachable from both — undecided by physics; grouped into regions and given to the side it shares the most **edge length** with.

The shared edge *is* the parting line that gets produced. Greater shared length gives the shorter core↔cavity interface — the principle a mold designer uses.

03 · PARTING LINE

Chain every core↔cavity edge end-to-end, tessellating curves **along the curve** — a circular rim becomes ~64 ordered points.

Verify closure, then score on six metrics — projected area 40 %, outer-boundary confidence 20 %, moldability 15 %.

Open chains ×**0.25** , so a broken loop can never win. Top two within 5 % → flagged ambiguous.

Part 1 closed · planar
edges · 69.5 mm · z = 15.00

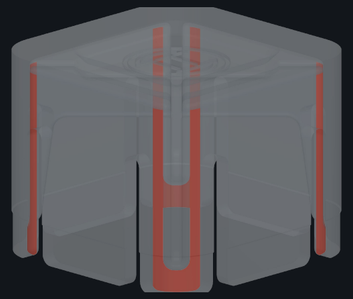
Part 3 closed · planar
circle ø36 · 113.1 mm · z = 4.00

core cavity line

What your feedback changed

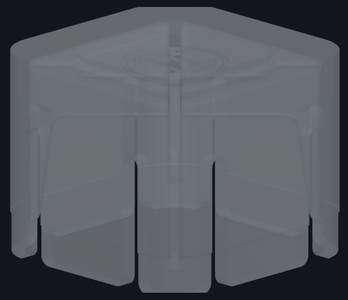
every item from the Phase 1 review call

JUNE



4 red undercut faces

AUGUST



none

4 → 0 undercuts on Part 1 — your reference answer

Score moved 87.2 → 88.0 and low-draft warnings 74 → 78. The extra four are real: draft is now the worst value across a face's samples, not the value at its midpoint.

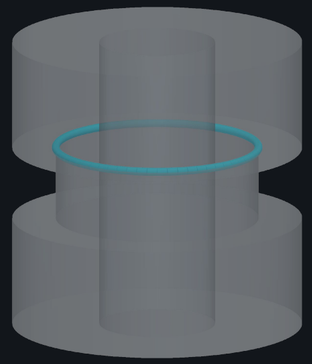
“Part 1 has zero undercuts — you reported four.”	UV-grid sampling, up to 15 points per face; only entering rays block a pull
“The parting line must be one closed continuous loop.”	Edges chained end-to-end, closure verified, open chains scored ×0.25
“Rank pull directions by undercut area, not count.”	Area-first ranking with branch-and-bound pruning — same winner, ~3× faster
“Even the internal surface is formed by the core.”	Reachability rays and the core-pin convention decide walls and through-holes
“A nozzle with an O-ring groove pulls perpendicular to its axis.”	Silhouette splitting along the clamshell plane — all three answers now tests
“An undercut is a tooling decision, not a failure.”	Trapped faces clustered into features, each with mechanism and axis
Our own gap: no way to catch a regression.	33-test suite plus a headless CLI that re-runs both reference parts

DEEP DIVE · YOUR REVIEW-CALL EXAMPLE

Testing the nozzle

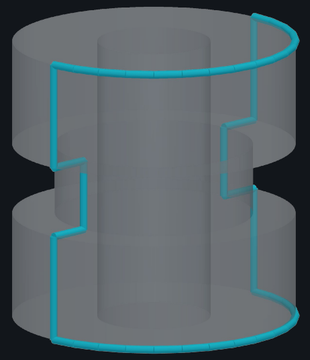
A turned nozzle with a bore and a 360° external O-ring groove. We rebuilt the part and tested for all three.

BEFORE



ACROSS the axis — cannot part

AFTER



ALONG the axis — clamshell

A groove that wraps the full circumference is trapped by any draw along the part's axis, at any parting-line height. An axial pull can never work.

PASS Pull runs **perpendicular** to the part axis

PASS Bore released by a **side core along the axis**

PASS Parting plane **contains** the part axis (clamshell)

What we had to build: silhouette splitting

A face whose normal flips across the pull is formed by **both** halves — but a classifier can only give a whole face to one half. On this nozzle **66 %** of the surface straddles; on Part 1, 0 %.

Fix: cut the **shell** — never the solid — along the plane containing the revolved axis. For a surface of revolution that one plane carries every silhouette, and it is exactly the clamshell plane.

guarded: skipped below 2 % straddling · skipped with no perpendicular revolved axis · falls back to the unsplit shape on failure

VALIDATION

Three independent checks

Your answer, an answer known by construction, and a mold that was actually built.

01 · YOUR ANSWERS

Your reference answers reproduce

Part 1 returns 0 undercuts and a flat closed rim. The nozzle reproduces all three answers you gave on the call.

Both are re-run headlessly by the CLI on every change.

02 · KNOWN BY CONSTRUCTION

33 tests, 32 passing, ~3 min

11 parts generated with cadquery — cup, capped cup, radial-hole cylinder, stepped cylinder, grooved cylinder, box, flanged boss, blind pocket, O-ring nozzle. The answer follows from how they were built: a cup's bore **must** belong to the core; a radial hole **must** be the only undercut under an axial pull. Nothing to argue with.

> Without the planarity assertion, Part 1's parting line drifted from a flat rim to a meandering loop across three commits and no test noticed.

03 · A MOLD THAT EXISTS

The plates meet at z = 0. We put the line at z = 0.00.

A public GrabCAD side-core mold ships its Cavity and Core Plates next to the parts they produce. PLASTIC BUSH matches at z = 0.00 with 2 undercut regions — the real mold uses a slide core. Coupler lands at z = 1.00; PLASTIC SLEEVE is weakest at z = 2.03.

On record from the validation log — the third-party CAD files are not redistributable, so that test skips itself on a clean clone.

RESULTS

Both reference parts, measured

METRIC	**Part 1** Phase 1 cap	**Part 3** Phase 2
Faces	311	414
Pull direction	**Z+**	**Z−**
Undercuts	**0**	88 faces = 2 regions, 1 axis
Core / cavity	269 / 42	322 / 4
Low-draft warnings	78	81
Primary parting line	closed rim, 8 edges, z = 15.00	closed circle ø36, z = 4.00
Ambiguous?	no	no
Manufacturability score	**88.0**	**81.8**
Time · UI path / exact ranking	**3.1 s** / 11.0 s	**21.5 s** / 44.9 s

Timings on a WSL2 laptop, single core, including STEP parse. The UI runs the pruned search first so the part renders quickly, then fills in the exact ranking from a second request — same winner either way.

SCALING TO PRODUCTION

How we'll make it scalable

One thing already done. Three levers left.

ALREADY DONE

Intersector built once per part and reused for every axis · axes not directions (±d symmetry) · branch-and-bound, biggest faces first.

PART 1

11.0 s → 3.1 s

PART 3

44.9 s → 21.5 s

Same winner, same score.

Deployment: one container, on-prem. No geometry leaves the network — no cloud call, no model, no telemetry.

THREE LEVERS LEFT

01	**Cache on file identity.**	Nothing is cached today, deliberately — per-face undercut flags are mutated in place during a sweep, so a shared cache would let one request corrupt another's. Make the sweep return results instead of tagging faces, then a hash of the STEP body keys the cache.
02	**The axis sweep is embarrassingly parallel.**	Every axis is an independent pass over the same geometry. The same fix unlocks both.
03	**Batch triage over a folder.**	The headless CLI already exists. Point it at a release folder overnight, get parts sorted by tooling complexity — that's the Bosch shape of this.

LIMITS & ROADMAP

Where the honest edges are

FOUR LIMITS

Silhouette splitting handles surfaces of revolution	A freeform part needing a non-planar split still gets a planar loop
One primary parting line per analysis	A stepped or multi-level parting surface is approximated by the closed loop
No wall thickness, flow, gate or cooling analysis	It is a moldability check, not a mold-flow simulation
Draft is flagged, not fixed	We do not modify geometry — no auto-drafting

ROADMAP, IN ORDER

01 Freeform (non-planar) silhouette splitting

02 Wall-thickness and sink-risk analysis

03 Stepped / multi-level parting surfaces

04 Multi-cavity layout, and a native CAD plugin so the check runs in-session

CLOSE

Where we got to, and how to run it

01 Every item from the Phase 1 review is closed, and each one is held closed by a test.

02 The part you drew on the call now returns all three of your own answers.

03 Undercuts read as tooling decisions — mechanism, axis and area — not as a face count.

```
git clone https://github.com/Vivek1537/dfm-agent.git
./app.sh# Windows: app.bat · then localhost:5173
```

WHAT SHIPS

Working tool · 4 documents · 33-test suite + CLI harness · two reference parts

THE ASK

Give us a part where you already know the answer and we do not. That is the fastest way to find where this breaks.

Vivek Boora · Ayush Pandey
Nitin · Afeera BMS College of Engineering